

Zewail City of Science and Technology

School of Computational Sciences and Artificial Intelligence (CSAI)

Graduation Project Final Submission Guidelines and Templates

Submission Deadline: June 20th

General Instructions

All submitted documents, presentations, posters, and videos must:

- Use a professional academic and technical writing style.
- Include the official Zewail City logo on the cover page or first slide.
- Clearly mention:
 - Project Title
 - Team Number
 - Supervisor Name
 - Student Names
 - Student IDs
 - Academic Programs
- Maintain consistent formatting, fonts, spacing, and citation style.
- Include proper references and citations for all external resources.
- Ensure originality and academic integrity.

Deliverables

1. Final Report / Thesis Template & Structure

The final report must contain the following chapters and sections.

Mandatory Content

Abstract (English and Arabic)

1. Introduction
2. Market Visibility and Business Case
3. Literature Review and Needed Background
4. System Design
5. Implementation Details
6. Testing and Evaluation
7. Ethics, Compliance, and Standards
8. Conclusion & Future Work
9. References
10. Appendix 1: User Guide
11. Appendix 2

Cover Page Requirements

All reports must clearly include:

- University logo
- Project title
- Supervisor name
- Student names
- Student IDs
- Academic programs
- Submission date

1. Thesis / Project Report Template



Zewail City of Science and Technology

School of Computational Sciences and Artificial Intelligence (CSAI)

Graduation Project Report

Project Title

Submitted by

Student Name	Student ID	Program
Student 1	ID	Program
Student 2	ID	Program
Student 3	ID	Program
Student 4	ID	Program

Supervisor

Dr. _____

Submitted in Partial Fulfillment of the Requirements for the Degree of

Bachelor of Science in CSAI

Date: June 2026

Formatting

- Font: Times New Roman
- Font Size:
 - Headings: 16–18
 - Body Text: 12
- Line Spacing: 1.5
- Margins: 2.5 cm
- Page Numbering Required
- Citation Style: IEEE or APA

Sample Introduction Structure

1.1 Background

Educational institutions increasingly rely on digital transformation technologies to improve operational efficiency, enhance student engagement, and automate administrative tasks. Traditional attendance systems suffer from several limitations including manual errors, proxy attendance, and time-consuming verification processes. Recent advancements in artificial intelligence and computer vision have enabled the development of automated attendance systems capable of identifying students using facial recognition and real-time video analytics.

Thesis Structure

Abstract

The report must contain:

- English abstract
- Arabic abstract

Both abstracts should summarize:

- Problem statement
- Motivation
- Proposed solution
- Main technologies used
- Evaluation approach
- Main findings and outcomes

Recommended length:

- 200–300 words per abstract

Table of Contents

Include:

- Chapters
- Sections
- Figures
- Tables
- Appendices

Chapter 1 — Introduction

This chapter should include:

- Final problem statement
- Background and motivation
- Proposed solution
- Project objectives
- Scope of the project
- Challenges addressed
- Contributions of the work
- Report organization

Chapter 2 — Market Visibility and Business Case

This chapter should discuss:

- Market relevance of the problem
- Target users and stakeholders
- Existing market gaps
- Competitive analysis
- Potential impact of the solution
- Innovation aspects
- Feasibility and sustainability
- Potential scalability and commercialization opportunities

Recommended:

- Include market analysis figures or statistics
- Compare with existing solutions
- Discuss business value and usability

Chapter 3 — Literature Review & Needed Background

Discuss:

- Existing systems and solutions
- Related research papers
- Commercial or industrial tools
- Comparative analysis
- Identified gaps and limitations
- Positioning of your work

Recommended:

- Use comparison tables
- Include references to recent work
- Explain how your project differs from existing approaches

Chapter 4 — System Design

This section should include:

Functional Requirements

- Main features
- User requirements
- Use cases

Data flow Design

Describe:

- Overall development approach
- Research methodology
- System workflow
- Algorithms and techniques
- Data collection and preprocessing (if applicable)
- AI/ML pipeline (if applicable)
- Development lifecycle followed

Non-Functional Requirements

Examples:

- Scalability
- Performance
- Security
- Reliability
- Maintainability
- Usability

Architecture Design

Include:

- High-level architecture diagram
- Context & Component diagrams
- Database design / ERD
- API design
- Deployment architecture
- Technology stack

UI/UX Design

Include:

- Wireframes
- Screenshots
- Navigation flow

Chapter 5 — Implementation Details

Describe the implementation details of the main modules.

This section should clearly discuss:

- Technologies and frameworks used
- Development tools and environments
- Key coding and engineering challenges
- How technical challenges were solved
- Important engineering decisions
- Optimization techniques
- Security and deployment considerations

The chapter should also include:

- Relevant code snippets
- Screenshots
- API examples
- Deployment examples
- Database implementation details (if applicable)

For each module include:

- Purpose
- Technologies/frameworks used
- Internal workflow
- Key implementation details
- Screenshots or code snippets

Examples:

- Authentication module
- AI/ML module
- Backend services
- Frontend application
- Database layer
- API Gateway
- Deployment infrastructure

Chapter 6 — Testing & Evaluation

Include:

Testing Types

Examples:

- Unit testing
- Integration testing
- System testing
- Performance testing
- Security testing
- Usability testing

Evaluation Metrics

Examples:

- Accuracy
- Precision/Recall/F1/SNR
- Response time
- Throughput
- Latency
- Resource utilization
- User satisfaction

Experimental Setup

Describe:

- Hardware/software environment
- Dataset details
- Configurations used
- Evaluation methodology

Chapter 7 — Results and Discussion

Present:

- Experimental results
- Graphs and charts
- Performance analysis
- Comparison with baselines
- Discussion of findings
- Interpretation of results
- Limitations

Recommended:

- Use figures and tables clearly
- Explain the significance of the results
- Discuss trade-offs and engineering decisions

Chapter 8 — Ethics, Compliance, and Standards

This chapter should consolidate the project's ethical, legal, compliance, and governance considerations.

Possible topics:

- Ethical AI considerations
- Data privacy and protection
- Security compliance
- Accessibility standards
- Responsible AI practices
- Sustainability considerations
- Regulatory requirements
- Risk mitigation strategies

Students should clearly explain how compliance and ethical engineering principles were integrated into the system design and implementation.

This chapter should clearly address:

Ethical Risks Assessed and Mitigated

Examples:

- Bias and fairness concerns
- Privacy risks
- Security threats
- AI hallucination risks
- Misuse scenarios
- Accessibility concerns
- Reliability and safety issues

Data Handling and Consent Mechanisms

Examples:

- User consent mechanisms
- Data collection policies
- Data storage and protection
- Data anonymization
- Access control mechanisms
- Compliance with privacy requirements

Standards Followed and Why

Examples:

- ISO standards
- OWASP guidelines
- GDPR principles
- Accessibility standards
- Secure software engineering standards
- AI governance guidelines

Students should explain:

- Which standards were selected
- Why they are relevant to the project
- How they were applied

Chapter 9 — Conclusions & Future work

This chapter should summarize:

- Main achievements
- Final outcomes
- Contributions
- Lessons learned
- How the project meets its objectives

- Possible future enhancements
- Additional features
- Scalability improvements
- Research extensions
- Long-term vision for the project

References

Requirements:

- Use consistent citation format
- Include all cited papers, tools, frameworks, datasets, and online resources
- Ensure references are properly formatted

IEEE Style Examples

Journal Paper

[1] J. Smith and A. Brown, “AI-based attendance management systems using facial recognition,” *Journal of Intelligent Systems*, vol. 15, no. 2, pp. 120–135, 2025.

Conference Paper

[2] M. Ali, S. Hassan, and R. Ibrahim, “Scalable cloud-native smart campus infrastructure,” in *Proceedings of the International Conference on Smart Computing*, Cairo, Egypt, 2024, pp. 88–95.

Book

[3] I. Sommerville, *Software Engineering*, 10th ed. Boston, MA, USA: Pearson, 2016.

Website

[4] OWASP Foundation, “OWASP Top 10 Web Application Security Risks,” 2025. [Online]. Available: <https://owasp.org/www-project-top-ten/>

Dataset

[5] Kaggle, “Face Recognition Dataset,” 2025. [Online]. Available: <https://www.kaggle.com/>

GitHub Repository

[6] OpenCV, “OpenCV Computer Vision Library,” GitHub Repository, 2025. [Online]. Available: <https://github.com/opencv/opencv>

Appendix 1 — User Guide

This appendix must provide a step-by-step guide explaining how to use the system.

The user guide should include:

- Installation instructions
- Setup instructions
- Configuration steps
- System requirements
- Deployment steps
- User workflows
- Screenshots and UI walkthroughs
- Troubleshooting tips
- Example use cases

The guide should be written clearly for both technical and non-technical users.

Appendix 2

Could include:

- Lessons learned
- Challenges encountered
- Potential improvements
- Future extensions

The appendices should include supporting materials such as:

- Project code samples
- Tools and frameworks used
- Detailed use cases
- Feasibility study
- Additional screenshots
- API documentation
- Survey forms
- Additional evaluation results
- Summary of GitHub repository activity
- Deployment configurations
- Dataset descriptions (if applicable)

2. Demo Video Requirements (3-4 Minutes)

Objective

The video should demonstrate:

- The problem addressed
- The proposed solution
- Key system features
- System workflow
- Main technical contribution
- Final working prototype

Recommended Video Structure

Time	Content
0:00 – 0:30	Problem statement and motivation
0:30 – 1:00	Project overview and architecture
1:00 – 2:30	Live system demo and features
2:30 – 3:00	Results, impact, and conclusion

Video Guidelines

- Maximum duration: 4 minutes
- Clear audio and readable screen recording
- Professional editing and transitions
- Avoid excessive animations or distractions
- Demonstrate actual functionality
- Include subtitles if possible
- Ensure stable and readable UI demonstrations

Recommended tools:

- OBS Studio

- Camtasia
- Canva
- PowerPoint Recording

3. GitHub Repository Final Submission

This is a checklist to ensure that the repository is complete, professional, reproducible, and clearly demonstrates the contribution of all team members.

GitHub Repository Requirements

Each team must submit the GitHub repository link containing:

Mandatory Repository Content

The repository must include:

- Finalized source code
- Final working build of the application
- README.md
- Setup instructions
- Deployment instructions
- User documentation
- Environment requirements
- API documentation
- Database schema
- Screenshots and sample outputs
- All required assets (images, videos, configuration files, etc.)
- Documentation
- Screenshots/sample outputs

README Template

Project Title

Short project description.

Team Members

Name	ID	Program
Student 1	ID	Program
Student 2	ID	Program

Supervisor

Dr. / Prof. _____

Problem Statement

Explain the problem being addressed.

Features

List the main system features.

System Architecture

Provide a high-level architecture overview.

Technologies Used

Examples:

- Frontend
- Backend
- Database

- AI/ML frameworks
- Cloud services
- DevOps tools

Setup Instructions

Provide step-by-step setup instructions.

Deployment Instructions

Explain how to deploy and run the system.

Usage Guide

Explain how to use the system.

Screenshots / Demo

Include screenshots or demo GIFs.

Repository Professionalism Requirements

The repository must demonstrate:

- Clean and maintainable code
- Logical folder organization
- Proper documentation
- Well-commented implementation
- Consistent naming conventions

- Meaningful commit messages
- Proper branching strategy (if applicable)
- Reproducible build and deployment process
- Clear project structure
- Meaningful commit messages
- Organized branches (if applicable)
- Proper documentation
- Clear contribution history from all team members

Important:

- All team members must contribute through Git commits.
- Commit history should clearly reflect the contribution of each student.
- Repositories with unclear contribution history may be questioned during the defense.
- The final repository should build and run successfully.
- All dependencies and assets must be correctly linked and documented.
- README instructions should allow evaluators to reproduce the project successfully.

4. Demo + Oral Defense

This is a live presentation and Q&A session.

Students must be prepared to explain:

- Design decisions
- System architecture
- Implementation details
- Evaluation methodology
- Engineering trade-offs
- Testing strategy
- Individual contributions

Presentation Outline

Section	Description
Introduction	Team members, project title, and overview
Problem Statement & Motivation	Explain the problem and importance
Proposed Solution	Present the implemented solution and features
System Design	Explain architecture and technologies used
Lessons Learned	Discuss challenges and gained experience
Future Work	Present possible future extensions
Live Demo	Demonstrate the system functionality
Q&A	Answer technical and design questions

5. Presentation Template (15 Minutes)

Presentation Structure

Slide	Content
1	Title Slide
2	Problem Statement
3	Motivation and Objectives
4	Literature Review / Existing Solutions
5	Proposed Solution
6	System Architecture
7	Methodology
8	Main Features
9	Technical Implementation
10	Testing and Evaluation
11	Results
12	Challenges and Lessons Learned
13	Conclusion
14	Future Work
15	Questions

Title Slide Template

Place University Logo Here

Project Title

Graduation Project Defense

Submitted by

Student Name	Student ID	Program
Student 1	ID	Program
Student 2	ID	Program

Supervisor

Dr. _____

School of Computational Sciences and Artificial Intelligence (CSAI)

Zewail City of Science and Technology, June 2026

Presentation Guidelines

- Duration: 15 minutes maximum
- Focus on technical contribution and engineering decisions
- Use diagrams and visuals effectively
- Avoid excessive text on slides
- Demonstrate system workflow clearly
- Ensure readability and professional design
- Practice time management before the defense

Recommended:

- Use architecture diagrams
- Show testing and evaluation results
- Explain technical challenges and solutions
- Include short demo clips if needed

5. Poster Template

Poster Sections

The poster should contain:

- Project Title
- Team Members
- Supervisor
- Problem Statement
- Objectives
- Proposed Solution
- System Architecture
- Methodology
- Results
- Technologies Used
- Conclusion
- QR Code for GitHub Repository or Demo

Poster Layout Recommendation

Header Section

- University logo
- Project title
- Team information
- Supervisor information

Main Body

Problem and Motivation

Short explanation of the problem.

Proposed Solution

Overview of the approach.

Architecture

Include diagrams.

Results

Include:

- Graphs
- Screenshots
- Evaluation metrics

Conclusion

Summarize contributions and impact.

Poster Design Guidelines

- Use large readable fonts
- Avoid excessive text
- Prioritize visuals and diagrams
- Use consistent formatting and colors
- Ensure high-resolution figures
- Keep the design clean and professional

Recommended Poster Size:

- A0 Portrait or Landscape

Final Submission Checklist

Before submission, ensure the following are completed:

Item	Completed
Thesis completed and reviewed	☐
Thesis follows required structure	☐
Demo video prepared	☐
GitHub repository finalized	☐
README completed	☐
Commit history verified	☐
Presentation finalized	☐
Poster finalized	☐
All student information added	☐
Supervisor name added	☐
References verified	☐
All files uploaded before deadline	☐

Important Notes

- All deliverables must be submitted before June 20th, at 11:59 pm.
- Late submissions may affect the defense process and will be penalized by 10% every day.
- Teams must be ready for technical questions during the defense.
- Students should be prepared to explain:
 - o Design decisions
 - o Implementation details
 - o Testing methodology
 - o Evaluation metrics
 - o Individual contributions

Best Practices for a Strong Graduation Project Submission

- Demonstrate engineering depth.
- Clearly justify architectural and technical decisions.
- Show evidence of testing and evaluation.
- Explain scalability, security, maintainability, and deployment considerations.
- Ensure the project can be reproduced using the provided setup instructions.
- Use professional diagrams and documentation.
- Focus on clarity, organization, and technical quality.

Contact and Announcements

The final defense schedule and additional announcements will be shared on Classroom.